Foundations for statistical inference - Sampling distributions

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In this lab, you will investigate the ways in which the statistics from a random sample of data can serve as point estimates for population parameters. We're interested in formulating a *sampling distribution* of our estimate in order to learn about the properties of the estimate, such as its distribution.

Setting a seed: We will take some random samples and build sampling distributions in this lab, which means you should set a seed at the start of your lab. If this concept is new to you, review the lab on probability.

Getting Started

Load packages

In this lab, we will explore and visualize the data using the **tidyverse** suite of packages. We will also use the **infer** package for resampling.

Let's load the packages.

```
library(tidyverse)
library(openintro)
library(infer)
```

The data

A 2019 Gallup report states the following:

> The premise that scientific progress benefits people has been embodied in discoveries throughout the ages – from the development of vaccinations to the explosion of technology in the past few decades, resulting in billions of supercomputers now resting in the hands and pockets of people worldwide. Still, not everyone around the world feels science benefits them personally.
>
> **Source:** World Science Day: Is Knowledge Power?

The Wellcome Global Monitor finds that 20% of people globally do not believe that the work scientists do benefits people like them. In this lab, you will assume this 20% is a true population proportion and learn about how sample proportions can vary from sample to sample by taking smaller samples from the population. We will first create our population assuming a population size of 100,000. This means 20,000 (20%) of the population think the work scientists do does not benefit them personally and the remaining 80,000 think it does.

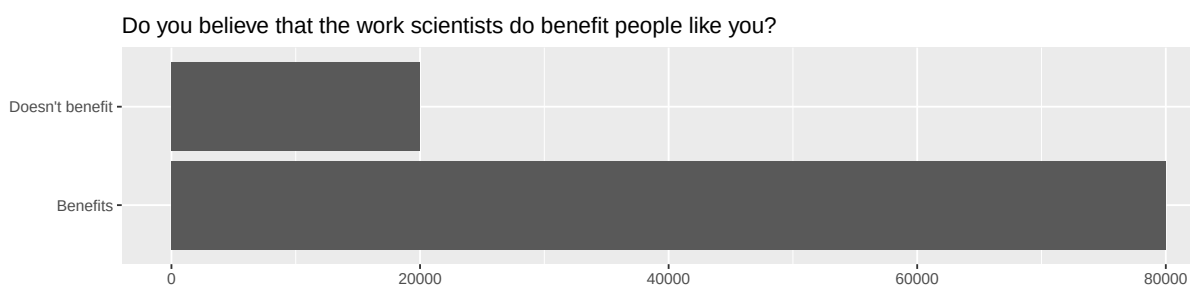
```
global_monitor <- tibble(
  scientist_work = c(rep("Benefits", 80000), rep("Doesn't benefit", 20000))
)
global_monitor
```

```
## # A tibble: 100,000 x 1
##    scientist_work
##    <chr>
##  1 Benefits
##  2 Benefits
##  3 Benefits
##  4 Benefits
##  5 Benefits
##  6 Benefits
##  7 Benefits
##  8 Benefits
##  9 Benefits
## 10 Benefits
## # i 99,990 more rows
```

The name of the data frame is `global_monitor` and the name of the variable that contains responses to the question *"Do you believe that the work scientists do benefit people like you?"* is `scientist_work`.

We can quickly visualize the distribution of these responses using a bar plot.

```
ggplot(global_monitor, aes(x = scientist_work)) +
  geom_bar() +
  labs(
    x = "", y = "",
    title = "Do you believe that the work scientists do benefit people like you?"
  ) +
  coord_flip()
```



We can also obtain summary statistics to confirm we constructed the data frame correctly.

```
global_monitor %>%
  count(scientist_work) %>%
  mutate(p = n /sum(n))
```

```
## # A tibble: 2 x 3
##   scientist_work      n     p
##   <chr>           <int> <dbl>
## 1 Benefits        80000   0.8
## 2 Doesn't benefit 20000   0.2
```

The unknown sampling distribution

In this lab, you have access to the entire population, but this is rarely the case in real life. Gathering information on an entire population is often extremely costly or impossible. Because of this, we often take

a sample of the population and use that to understand the properties of the population.

If you are interested in estimating the proportion of people who don't think the work scientists do benefits them, you can use the `sample_n` command to survey the population.

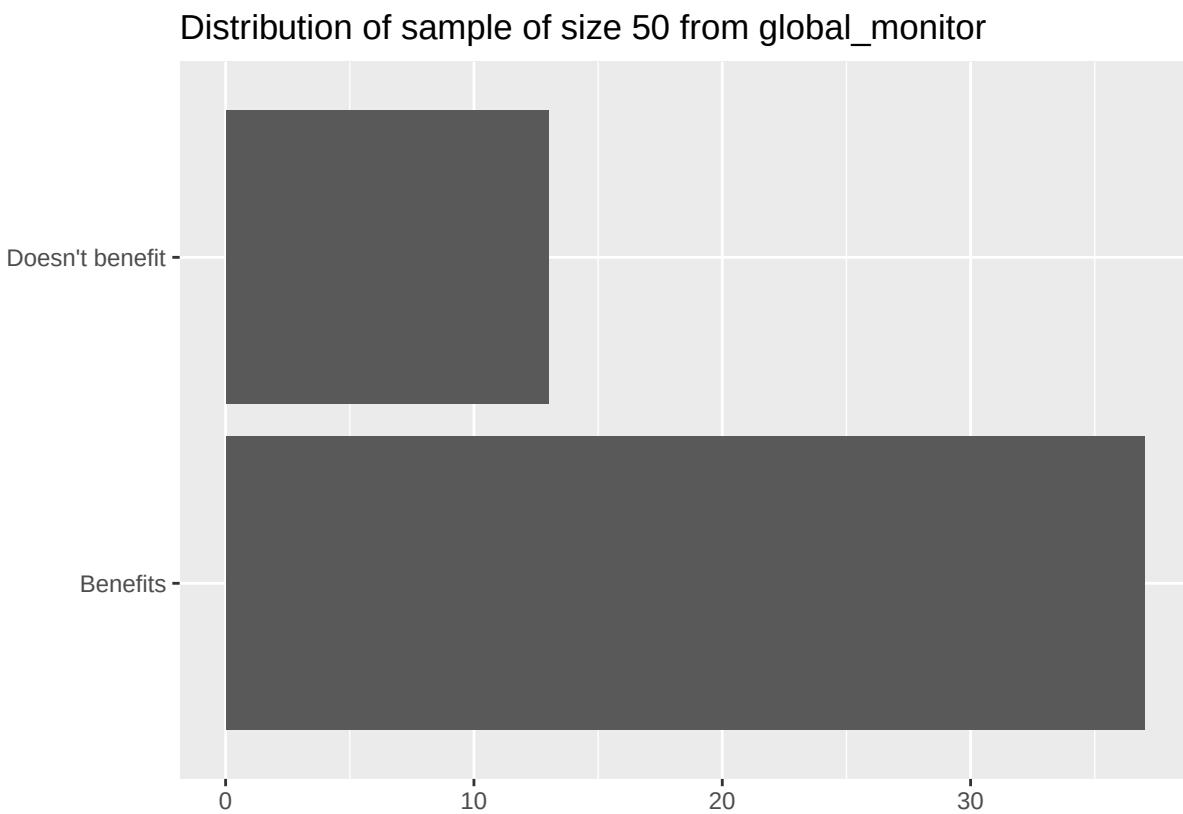
```
samp1 <- global_monitor %>%
  sample_n(50)
```

This command collects a simple random sample of size 50 from the `global_monitor` dataset, and assigns the result to `samp1`. This is similar to randomly drawing names from a hat that contains the names of all in the population. Working with these 50 names is considerably simpler than working with all 100,000 people in the population.

1. Describe the distribution of responses in this sample. How does it compare to the distribution of responses in the population. **Hint:** Although the `sample_n` function takes a random sample of observations (i.e. rows) from the dataset, you can still refer to the variables in the dataset with the same names. Code you presented earlier for visualizing and summarizing the population data will still be useful for the sample, however be careful to not label your proportion `p` since you're now calculating a sample statistic, not a population parameters. You can customize the label of the statistics to indicate that it comes from the sample.

Insert your answer here Visualization

```
ggplot( samp1, aes(x = scientist_work)) +
  geom_bar() +
  labs(
    x = "", y = "",
    title = "Distribution of sample of size 50 from global_monitor"
  ) +
  coord_flip()
```



Summarize: The distribution of responses in the sample of size 50 shows that most respondents believe that the work scientists do benefits them, while a smaller portion believe that it does not. This pattern is similar to the population distribution, where 80% believe science benefits them and 20% do not. However, because the sample is randomly selected and much smaller than the population, the proportions in the sample may differ slightly from the population proportions.

If you're interested in estimating the proportion of all people who do not believe that the work scientists do benefits them, but you do not have access to the population data, your best single guess is the sample mean.

```
samp1 %>%
  count(scientist_work) %>%
  mutate(p_hat = n /sum(n))
```

```
## # A tibble: 2 x 3
##   scientist_work      n p_hat
##   <chr>           <int> <dbl>
## 1 Benefits           37  0.74
## 2 Doesn't benefit    13  0.26
```

Depending on which 50 people you selected, your estimate could be a bit above or a bit below the true population proportion of 0.26. In general, though, the sample proportion turns out to be a pretty good estimate of the true population proportion, and you were able to get it by sampling less than 1% of the population.

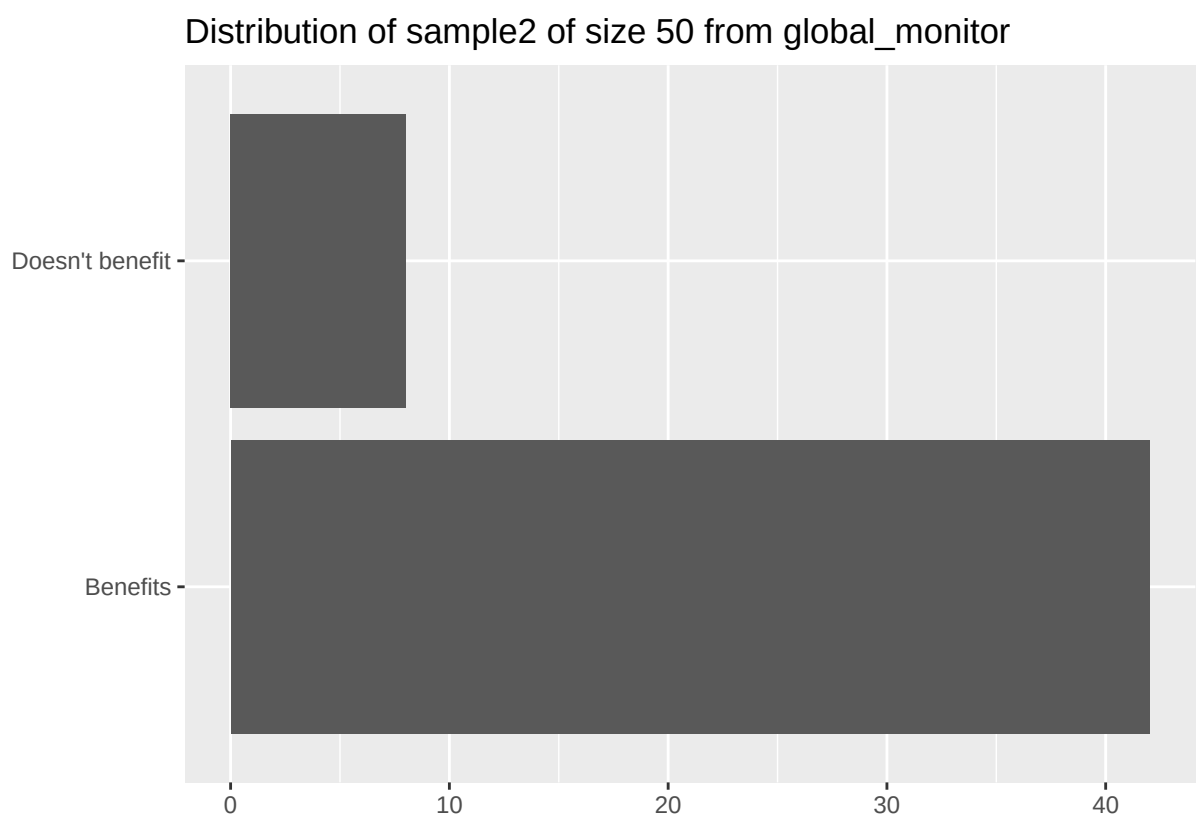
2. Would you expect the sample proportion to match the sample proportion of another student's sample? Why, or why not? If the answer is no, would you expect the proportions to be somewhat different or very different? Ask a student team to confirm your answer.

Insert your answer here: Answer** **I would not expect the sample proportion to exactly match another student's sample proportion because each sample is selected randomly from the population. Different random samples may contain different individuals, which leads to different proportions. However, the proportions should not be drastically different and should generally be close to the true population proportion.** **

3. Take a second sample, also of size 50, and call it `samp2`. How does the sample proportion of `samp2` compare with that of `samp1`? Suppose we took two more samples, one of size 100 and one of size 1000. Which would you think would provide a more accurate estimate of the population proportion?

Insert your answer here visualizing: a sample of size 50 and call it samp2

```
samp2 <- global_monitor %>%
  sample_n(50)
ggplot( samp2, aes(x = scientist_work)) +
  geom_bar() +
  labs(
    x = "", y = "",
    title = "Distribution of sample2 of size 50 from global_monitor"
  ) +
  coord_flip()
```




```
samp2 %>%
  count(scientist_work) %>%
  mutate(p_hat2 = n / sum(n))
```

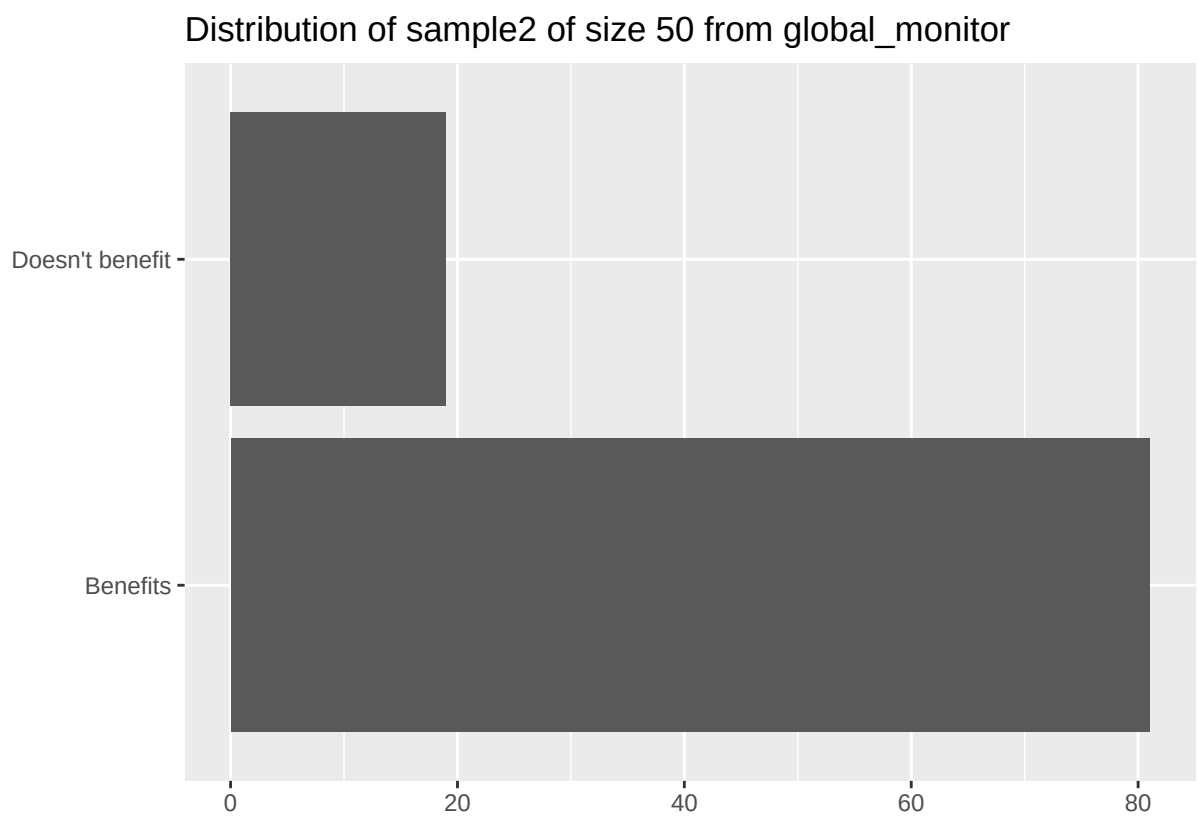
```
## # A tibble: 2 x 3
##   scientist_work      n p_hat2
##   <chr>           <int>  <dbl>
## 1 Benefits           42   0.84
## 2 Doesn't benefit     8   0.16
```

Summarizing: with size 50 of samp2 we can see that 39 says it benefits them while 11 says it doesn't benefit them which is different from samp1

Using a size of 100 and comparing with a size of 1000

sample of 100

```
samp_100 <- global_monitor %>%
  sample_n(100)
ggplot( samp_100, aes(x = scientist_work)) +
  geom_bar() +
  labs(
    x = "", y = "",
    title = "Distribution of sample2 of size 50 from global_monitor"
  ) +
  coord_flip()
```

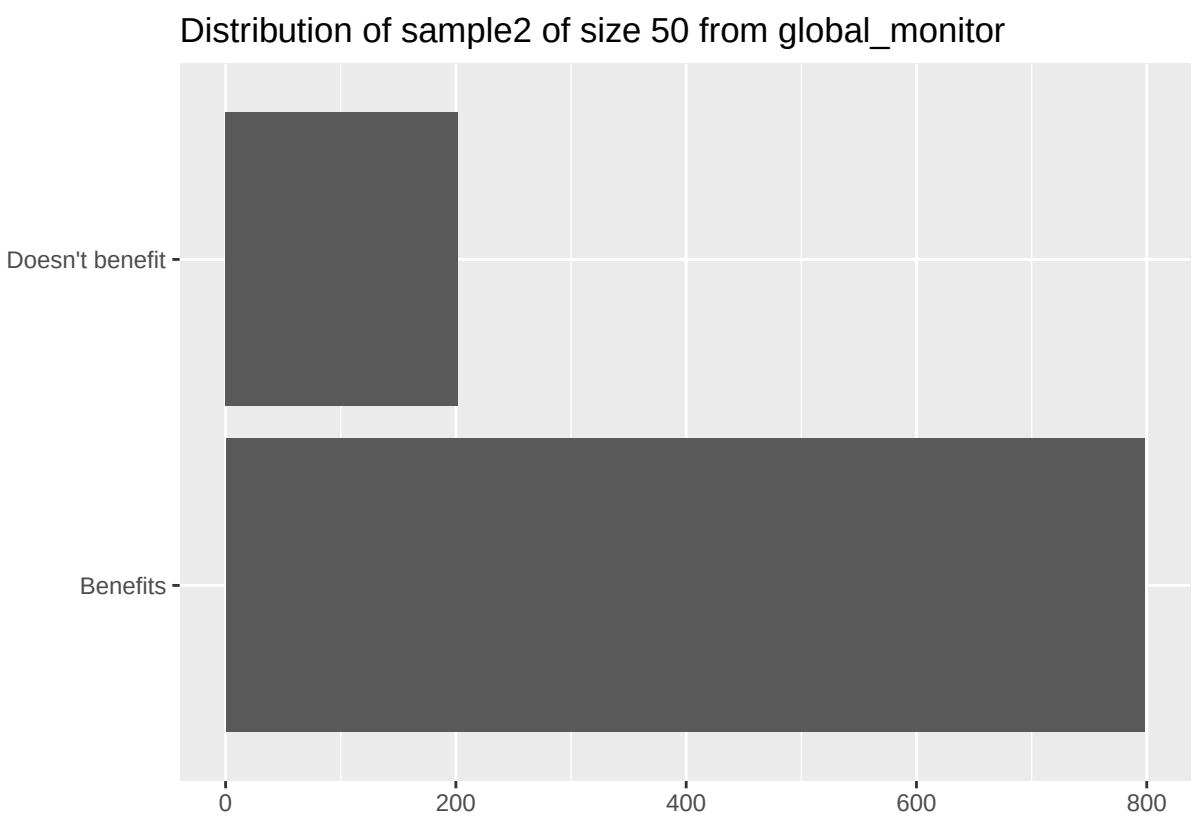



```
samp_100 %>%
  count(scientist_work) %>%
  mutate(p_hat_100 = n / sum(n))
```

```
## # A tibble: 2 x 3
##   scientist_work      n p_hat_100
##   <chr>           <int>     <dbl>
## 1 Benefits           81      0.81
## 2 Doesn't benefit    19      0.19
```

Using sample of 1000

```
samp_1000 <- global_monitor %>%
  sample_n(1000)
ggplot( samp_1000, aes(x = scientist_work)) +
  geom_bar() +
  labs(
    x = "", y = "",
    title = "Distribution of sample2 of size 50 from global_monitor"
  ) +
  coord_flip()
```




```
samp_1000 %>%
  count(scientist_work) %>%
  mutate(p_hat_1000 = n / sum(n))
```

```
## # A tibble: 2 x 3
##   scientist_work      n p_hat_1000
##   <chr>           <int>      <dbl>
```

```
## 1 Benefits          798      0.798
## 2 Doesn't benefit   202      0.202
```

Answer: The sample proportion from samp2 may differ from the proportion in samp1 because each sample is drawn randomly from the population. Random variation can cause the estimates to differ slightly. If we took additional samples of size 100 and 1000, the sample of size 1000 would provide a more accurate estimate of the population proportion. Larger sample sizes reduce sampling variability and produce estimates that are closer to the true population proportion.*

Not surprisingly, every time you take another random sample, you might get a different sample proportion. It's useful to get a sense of just how much variability you should expect when estimating the population mean this way. The distribution of sample proportions, called the *sampling distribution (of the proportion)*, can help you understand this variability. In this lab, because you have access to the population, you can build up the sampling distribution for the sample proportion by repeating the above steps many times. Here, we use R to take 15,000 different samples of size 50 from the population, calculate the proportion of responses in each sample, filter for only the *Doesn't benefit* responses, and store each result in a vector called `sample_props50`. Note that we specify that `replace = TRUE` since sampling distributions are constructed by sampling with replacement.

```r
sample_props50 <- global_monitor %>%
                    rep_sample_n(size = 50, reps = 15000, replace = TRUE) %>%
                    count(scientist_work) %>%
                    mutate(p_hat = n /sum(n)) %>%
                    filter(scientist_work == "Doesn't benefit")
```

And we can visualize the distribution of these proportions with a histogram.

```r
ggplot(data = sample_props50, aes(x = p_hat)) +
  geom_histogram(binwidth = 0.02) +
  labs(
    x = "p_hat (Doesn't benefit)",
    title = "Sampling distribution of p_hat",
    subtitle = "Sample size = 50, Number of samples = 15000"
  )
```

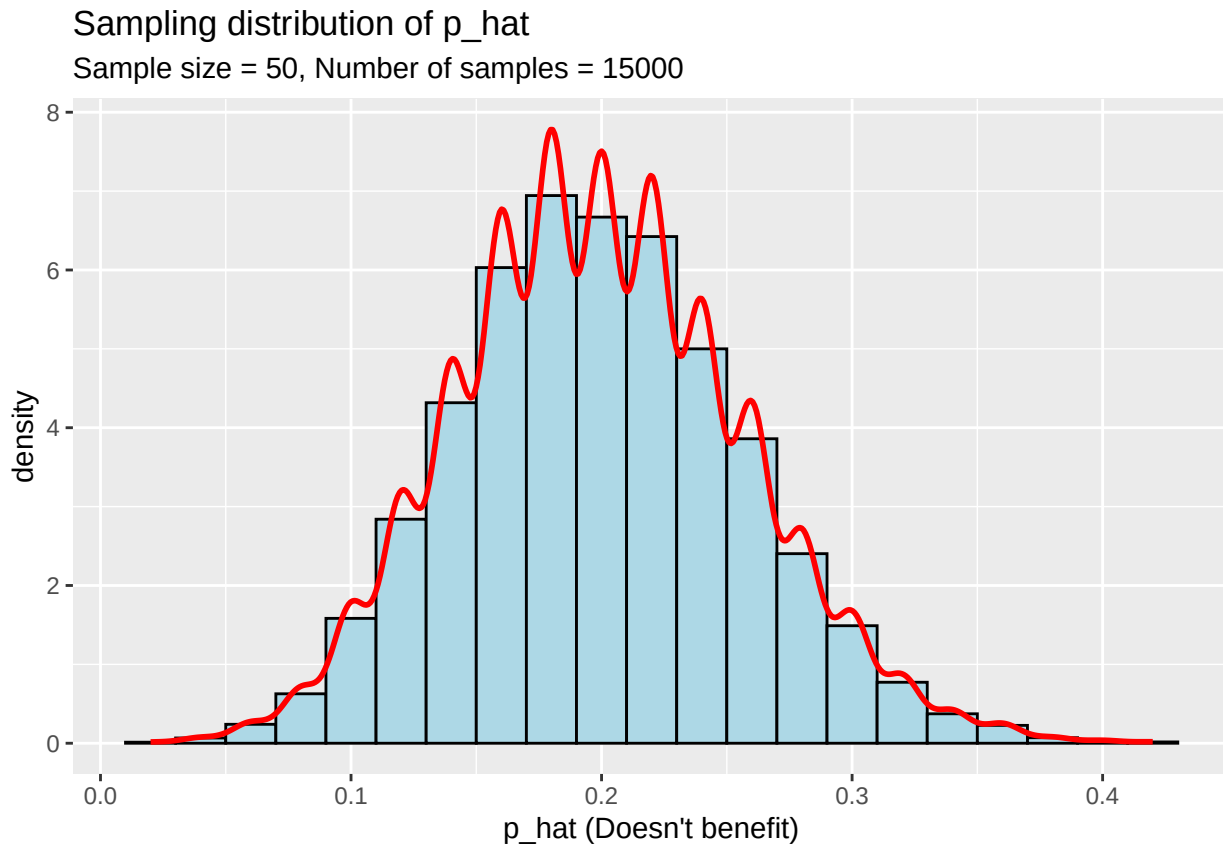
Next, you will review how this set of code works.

4. How many elements are there in `sample_props50`? Describe the sampling distribution, and be sure to specifically note its center. Make sure to include a plot of the distribution in your answer.

Insert your answer here: Answer: There are 15,000 elements in sample_props50, corresponding to the 15,000 simulated samples taken from the population. Each element represents the sample proportion of respondents who answered "Doesn't benefit" in a sample of size 50. The sampling distribution of the sample proportions appears approximately bell-shaped and symmetric, resembling a normal distribution. The center of the distribution is around 0.20, which matches the true population proportion of individuals who believe that the work scientists do does not benefit them.**

```r
ggplot(sample_props50, aes(x = p_hat)) +
  geom_histogram(aes(y = ..density..), binwidth = 0.02, fill = "lightblue", color = "black") +
  geom_density(color = "red", size = 1) +
  labs(
```

```
    x = "p_hat (Doesn't benefit)",
    title = "Sampling distribution of p_hat",
    subtitle = "Sample size = 50, Number of samples = 15000"
  )
```



Interlude: Sampling distributions

The idea behind the `rep_sample_n` function is *repetition*. Earlier, you took a single sample of size `n` (50) from the population of all people in the population. With this new function, you can repeat this sampling procedure `rep` times in order to build a distribution of a series of sample statistics, which is called the **sampling distribution**.

Note that in practice one rarely gets to build true sampling distributions, because one rarely has access to data from the entire population.

Without the `rep_sample_n` function, this would be painful. We would have to manually run the following code 15,000 times

```
global_monitor %>%
  sample_n(size = 50, replace = TRUE) %>%
  count(scientist_work) %>%
  mutate(p_hat = n /sum(n)) %>%
  filter(scientist_work == "Doesn't benefit")
```

```
## # A tibble: 1 x 3
```

```
##   scientist_work      n p_hat
##   <chr>           <int> <dbl>
## 1 Doesn't benefit    11  0.22
```

as well as store the resulting sample proportions each time in a separate vector.

Note that for each of the 15,000 times we computed a proportion, we did so from a **different** sample!

5. To make sure you understand how sampling distributions are built, and exactly what the `rep_sample_n` function does, try modifying the code to create a sampling distribution of **25 sample proportions** from **samples of size 10**, and put them in a data frame named `sample_props_small`. Print the output. How many observations are there in this object called `sample_props_small`? What does each observation represent?

Insert your answer here: Answer: we have 25 observation are in this sample_props_small and each observation the number of percentage in each sample of size 10 the number of people that Doesn't benefit from science.

```
sample_props_small <- global_monitor %>%
  rep_sample_n(size =  10, reps = 25, replace = TRUE) %>%
  count(scientist_work) %>%
  mutate( p_hat = n / sum(n)) %>%
  filter(scientist_work == "Doesn't benefit")
sample_props_small
```

```
## # A tibble: 22 x 4
## # Groups:   replicate [22]
##    replicate scientist_work      n p_hat
##        <int> <chr>           <int> <dbl>
##  1         1 Doesn't benefit     2   0.2
##  2         2 Doesn't benefit     4   0.4
##  3         3 Doesn't benefit     5   0.5
##  4         5 Doesn't benefit     2   0.2
##  5         6 Doesn't benefit     1   0.1
##  6         7 Doesn't benefit     5   0.5
##  7         9 Doesn't benefit     1   0.1
##  8        10 Doesn't benefit     1   0.1
##  9        11 Doesn't benefit     2   0.2
## 10        12 Doesn't benefit     4   0.4
## # i 12 more rows
```

Sample size and the sampling distribution

Mechanics aside, let's return to the reason we used the `rep_sample_n` function: to compute a sampling distribution, specifically, the sampling distribution of the proportions from samples of 50 people.

```
ggplot(data = sample_props50, aes(x = p_hat)) +
  geom_histogram(binwidth = 0.02)
```

The sampling distribution that you computed tells you much about estimating the true proportion of people who think that the work scientists do doesn't benefit them. Because the sample proportion is an unbiased estimator, the sampling distribution is centered at the true population proportion, and the spread of the

distribution indicates how much variability is incurred by sampling only 50 people at a time from the population.

In the remainder of this section, you will work on getting a sense of the effect that sample size has on your sampling distribution.

6. Use the app below to create sampling distributions of proportions of *Doesn't benefit* from samples of size 10, 50, and 100. Use 5,000 simulations. What does each observation in the sampling distribution represent? How does the mean, standard error, and shape of the sampling distribution change as the sample size increases? How (if at all) do these values change if you increase the number of simulations? (You do not need to include plots in your answer.)

Insert your answer here: Answer: Each observation in the sampling distribution represents the sample proportion of people who said "Doesn't benefit" in one simulated sample. As the sample size increases from 10 to 50 to 100, the mean of the sampling distribution stays close to the true population proportion of 0.20, while the standard error decreases. This means the distribution becomes less spread out and the estimates become more precise. In addition, the shape becomes more nearly bell-shaped as the sample size increases. If the number of simulations is increased, the overall shape becomes smoother and more stable, but the center and standard error do not change much

---

More Practice

So far, you have only focused on estimating the proportion of those you think the work scientists doesn't benefit them. Now, you'll try to estimate the proportion of those who think it does.

Note that while you might be able to answer some of these questions using the app, you are expected to write the required code and produce the necessary plots and summary statistics. You are welcome to use the app for exploration.

7. Take a sample of size 15 from the population and calculate the proportion of people in this sample who think the work scientists do enhances their lives. Using this sample, what is your best point estimate of the population proportion of people who think the work scientists do enchances their lives?

Insert your answer here Answer: using this sample the best point estimate of people who think science enhance their lives is almost 92% of the population

```
sample_1 <- global_monitor %>%
  sample_n(size = 15, replace = TRUE) %>%
  count(scientist_work) %>%
  mutate(p_hat = n /sum(n)) %>%
  filter(scientist_work == "Benefits")
sample_1
```

```
## # A tibble: 1 x 3
##   scientist_work     n p_hat
##   <chr>          <int> <dbl>
## 1 Benefits          15     1
```

8. Since you have access to the population, simulate the sampling distribution of proportion of those who think the work scientists do enchances their lives for samples of size 15 by taking 2000 samples from the population of size 15 and computing 2000 sample proportions. Store these proportions in as `sample_props15`. Plot the data, then describe the shape of this sampling distribution. Based on this sampling distribution, what would you guess the true proportion of those who think the work scientists do enchances their lives to be? Finally, calculate and report the population proportion.

Insert your answer here Answer: the shape of this sample distribution of the sample proportions appears approximately bell-shaped and symmetric, resembling a normal distribution. The true proportion of those who think the work of science do enchances their lives is aproximatly 0.80 or 80%

```
global_monitor %>%
  summarise(p = mean(scientist_work == "Benefits"))
```

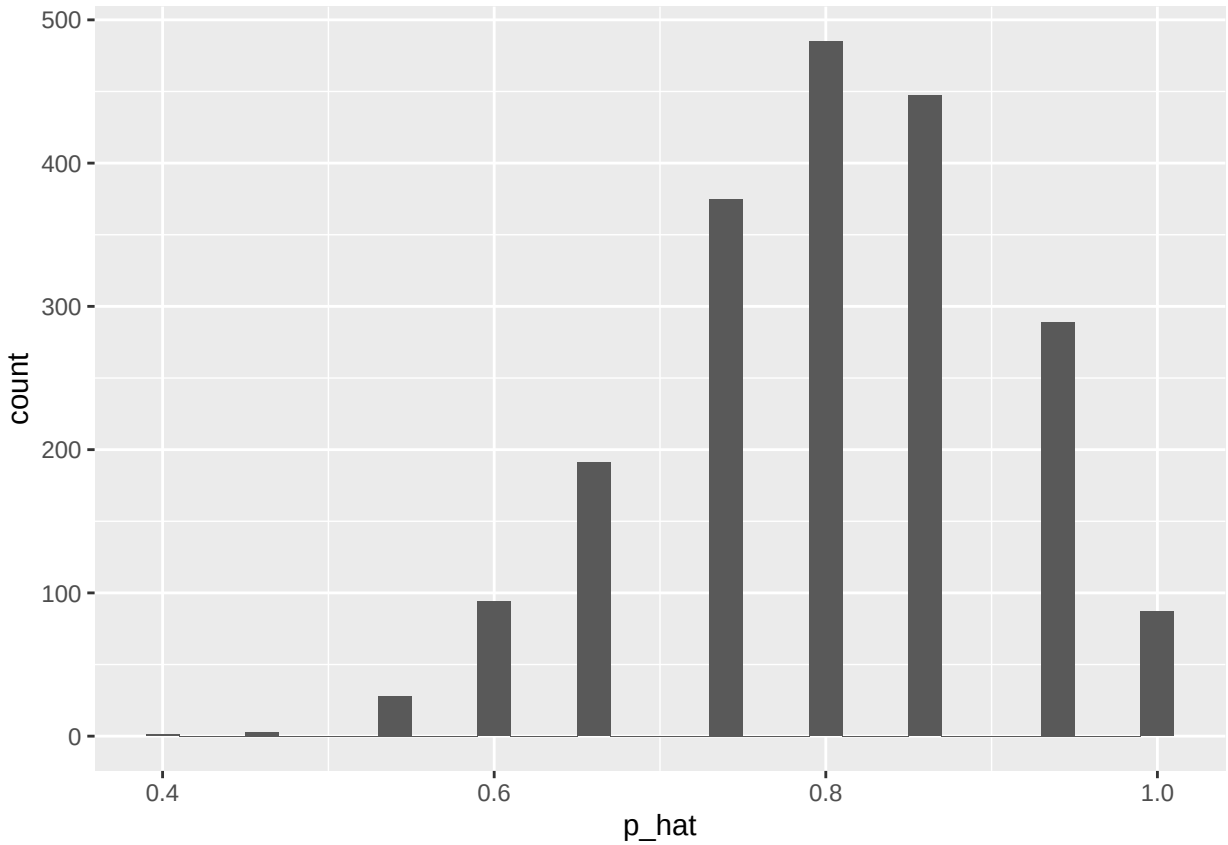
```
## # A tibble: 1 x 1
##       p
##   <dbl>
## 1   0.8
```

```
sample_props15 <- global_monitor %>%
  rep_sample_n(size = 15, reps = 2000, replace = TRUE) %>%
  count(scientist_work) %>%
  mutate(p_hat = n /sum(n)) %>%
  filter(scientist_work == "Benefits")
sample_props15
```

```
## # A tibble: 2,000 x 4
## # Groups:   replicate [2,000]
##    replicate scientist_work     n p_hat
##        <int> <chr>          <int> <dbl>
##  1         1 Benefits          13 0.867
##  2         2 Benefits          10 0.667
##  3         3 Benefits          14 0.933
##  4         4 Benefits          13 0.867
##  5         5 Benefits          11 0.733
##  6         6 Benefits          14 0.933
##  7         7 Benefits          13 0.867
##  8         8 Benefits          14 0.933
##  9         9 Benefits          10 0.667
## 10        10 Benefits          13 0.867
## # i 1,990 more rows
```

Using the plot

```
ggplot(data = sample_props15, aes(x = p_hat)) +
  geom_histogram(binwidth = 0.02)
```



9. Change your sample size from 15 to 150, then compute the sampling distribution using the same method as above, and store these proportions in a new object called `sample_props150`. Describe the shape of this sampling distribution and compare it to the sampling distribution for a sample size of 15. Based on this sampling distribution, what would you guess to be the true proportion of those who think the work scientists do enchances their lives?

Insert your answer here Answer:The sampling distribution of the sample proportions appears approximately bell-shaped and symmetric, resembling a normal distribution. The center of the distribution is around 0.80, which matches the true population proportion of individuals who believe that the work scientists do does benefits them. we can estimate the proportion to be around 80%

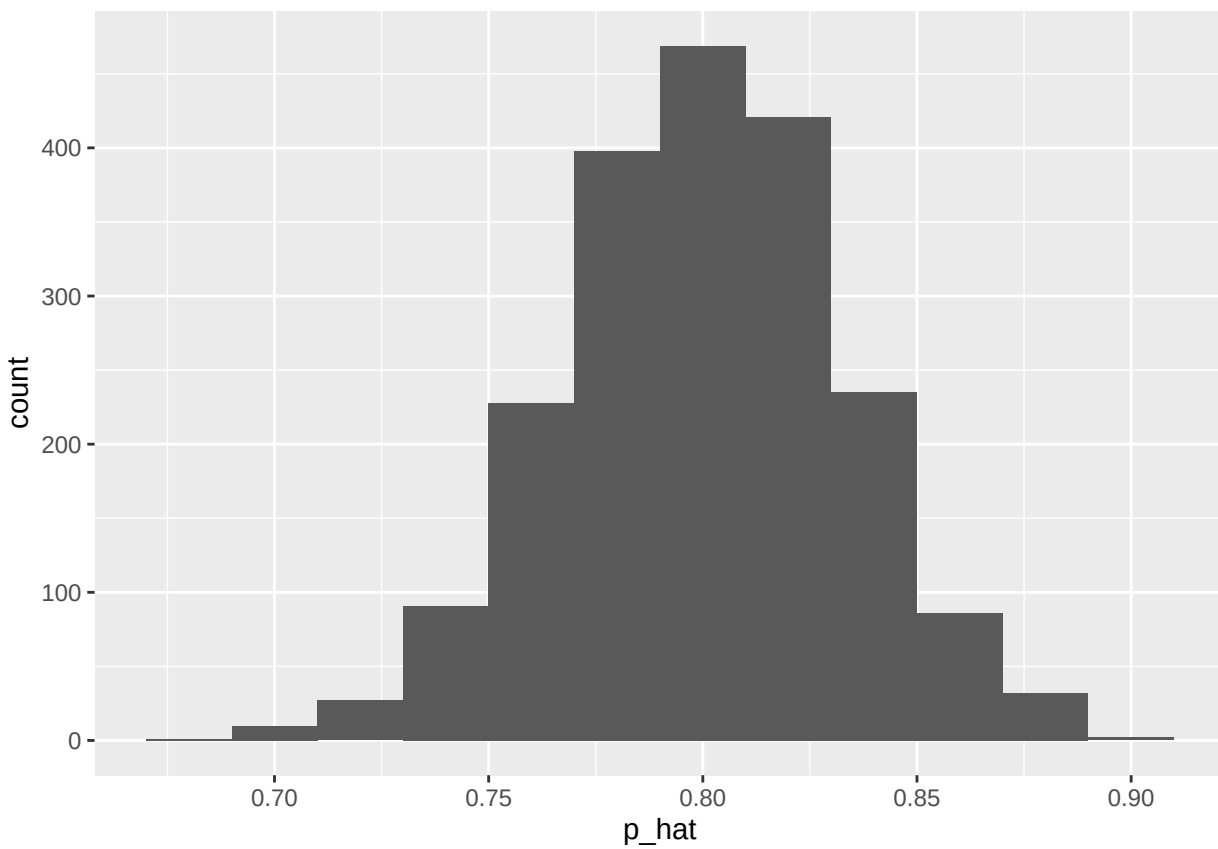
```
sample_props150 <- global_monitor %>%
  rep_sample_n(size = 150, reps = 2000, replace = TRUE) %>%
  count(scientist_work) %>%
  mutate(p_hat = n /sum(n)) %>%
  filter(scientist_work == "Benefits")
sample_props150
```

```
## # A tibble: 2,000 x 4
## # Groups:   replicate [2,000]
##    replicate scientist_work     n p_hat
##        <int> <chr>          <int> <dbl>
##  1         1 Benefits         120 0.8
##  2         2 Benefits         127 0.847
##  3         3 Benefits         122 0.813
##  4         4 Benefits         116 0.773
```

```
## 5          5 Benefits         123 0.82
## 6          6 Benefits         122 0.813
## 7          7 Benefits         126 0.84
## 8          8 Benefits         119 0.793
## 9          9 Benefits         120 0.8
## 10        10 Benefits         114 0.76
## # i 1,990 more rows
```

Using the plot

```
ggplot(data = sample_props150, aes(x = p_hat)) +
  geom_histogram(binwidth = 0.02)
```



10. Of the sampling distributions from 2 and 3, which has a smaller spread? If you're concerned with making estimates that are more often close to the true value, would you prefer a sampling distribution with a large or small spread? **Answer: The sampling distribution based on the sample size of 150 has the smaller spread. If I want estimates that are more often close to the true population proportion, I would prefer a sampling distribution with a small spread, because that indicates less variability and more precise estimates**